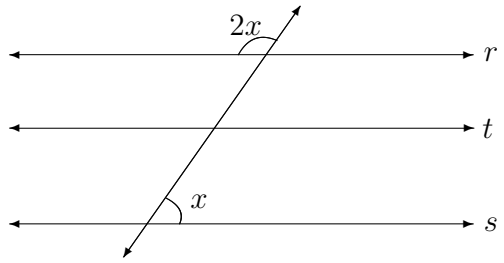
## NMC 2011 PRIMARY LEVEL PAPER 2 QUESTIONS

### SECTION A : 5 marks each

Qn. 1. Which of the following numbers is the largest:

$$9.12344, \quad 9.123\overline{4}, \quad 9.12\overline{34}, \quad 9.1\overline{234}, \quad 9.\overline{1234}\,?$$ _____

Qn. 2. Lines $r$, $t$ and $s$ are parallel, angle $x$ equals _____



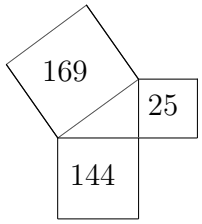

Qn. 3. A 27 year old mother has a 5 year old daughter. In how many years will the mother be three times as old as her daughter?

Qn. 4. A mixture of 30 litres of paint is 25% red paint, 30% yellow tint and 45% water. Five litres of yellow tint are added to the original mixture. What is the percent of yellow tint in the new mixture?

Qn. 5. Which of these five numbers is closest to $4.7$ ?
$$4.5, \quad 4.8, \quad 4.62, \quad 4.7264, \quad 4.68834$$

Qn. 6. Joseph had walked half way from home to school when he realised that he was late. He ran the rest of the way to school. He ran 3 times as fast as he walked. Joseph took 6 minutes to walk half way to school. How many minutes did it take Joseph to get from home to school?

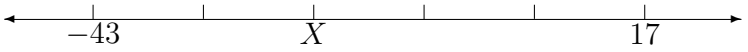
Qn. 7. Given the areas (in $\text{cm}^2$) of the three squares in the figure, what is the area of the interior triangle?



Qn. 8. Farida's parents have agreed to pay for her trip if she spends an average of 10 hours per week helping around the house for 6 weeks. For the first 5 weeks, she helps around for $8, 11, 7, 12$ and 10 hours respectively. How many hours must she work during the final week to earn the trip?

Qn. 9. Use the digits $1, 2, 3, 4, 9$ to form the smallest possible even five-digit number.

Qn. 10. The markings on this number line are equally spaced. What is the value of $X$?

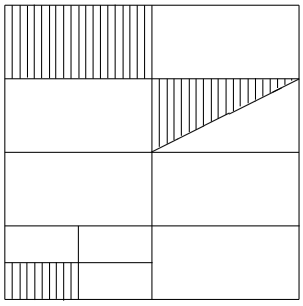



**SECTION B :** 10 **marks each**

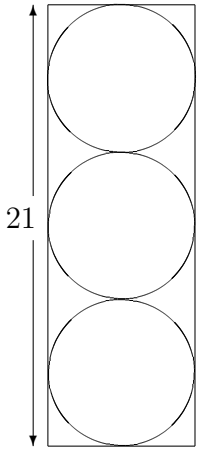
Qn. 11. Complete the grid below, each of the digits 1 through 4 must occur once in each row and once in each column.

| | | | |
|---|---|---|---|
| 1 | | 2 | |
| 2 | 3 | | |
| | | | 4 |
| | | | |

Qn. 12. What fraction of this square is shaded?



Qn. 13. The three circles touch the rectangle and touch each other as shown in the figure below. The length of the rectangle is 21cm. Find the area inside the rectangle but outside the circles (take $\pi = \frac{22}{7}$).




Qn. 14. Annet began peeling a pile of 44 bananas (matoke) at the rate of 3 bananas per minute. Four minutes later, Aisha joined her and peeled at the rate of 5 bananas per minute.

(a) When they finished, how many bananas had Aisha peeled?

(b) How long would it have taken them to peel all the 44 bananas if they had started at the same time?

Qn. 15. Ms. Kalibbala asks each pupil in her class to draw a rectangle whose sides are whole numbers and whose perimeter is 50cm. When marking the pupils work, she discovers that the pupils drew all the different possible rectangles.

(a) How many distinct (different) rectangles do the pupils come up with?

(b) All her pupils calculate the area of the rectangle they draw. What is the difference between the largest and smallest possible areas of the rectangles?